

one or more bitmaps showing the depths of the scene from the light's point of view. 3ds Max will then compare these bitmaps with the objects in the scene to determine which areas are in the shadow. To control this process, 3ds Max provides a Shadow Map Params rollout to adjust these parameters.

The two most important parameters to understand when working with depth map shadows are the Size and the Sample Range parameters. The Size parameter determines the size of the shadow map, which covers the width of the light cone. The Sample Range parameter controls the soft edge of the shadow. The larger the size of the shadow map, the more accurate the shadow. The larger the sample range, the more softening. Softening, however, is relative to the resolution, so larger size maps require more filtering to get the same softness. Although large maps are more accurate, they take up more memory when rendering. The size of the map depends on the cone angle of the light. If the light covers an area the size of the image, the depth map should be approximately the size of the image. Sometimes smaller maps are preferred, because they can produce softer shadows with less filtering.

There are other parameters you can set in this rollout. Two sided shadows will render shadows for both sides of a surface. Bias is a parameter that offsets the shadow from the object by a small amount to ensure that the shadow is behind the object. If the parameter is too low, parts of the shadow will render in front of the object. If it is too high, the shadow will be offset from the object.

### Raytraced Shadows

Raytraced shadows work by tracing the actual beams of light through the scene. This is more computationally expensive than simple depth map shadows, but raytraced shadows prove to be more accurate. They can create shadows for transparent objects, and they are also easier to set up, because you do not have to calculate map sizes. 3ds Max has two types of raytraced shadows: regular and advanced. The Advanced Raytraced Shadows option has additional parameters for controlling the look of the shadow.

The Ray Traced Shadow Params rollout has options to control the bias of the shadow, which is very similar to the bias control on depth map shadows. You can also use this rollout to turn on two-sided shadows.

### Advanced Raytraced Shadows

Advanced raytraced shadows are an improvement on standard raytraced shadows. These shadows have additional parameters to control the